```
restart;
 n := 4 :
result := convert(mtaylor(u(x + p·h, t + q·τ), [h = 0, τ = 0], n), diff) + O(h^4) + O(τ^4) + O(h
    ·τ^3) + O(h^2τ^2) + O(h^3·τ)
```

$$result := u(x,t) + \left(\frac{\partial}{\partial x}\,u(x,t)\right) p\,h + \left(\frac{\partial}{\partial t}\,u(x,t)\right) q\,\tau + \left(\frac{\partial^2}{\partial t \partial x}\,u(x,t)\right) h\,p\,q\,\tau \tag{1}$$
$$+ \frac{\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right) q^2\,\tau^2}{2} + \frac{\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right) h^2\,p^2}{2} + \frac{\left(\frac{\partial^3}{\partial t \partial x^2}\,u(x,t)\right) h^2\,p^2\,q\,\tau}{2}$$
$$+ \frac{\left(\frac{\partial^3}{\partial t^2 \partial x}\,u(x,t)\right) h\,p\,q^2\,\tau^2}{2} + \frac{\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right) q^3\,\tau^3}{6} + \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right) h^3\,p^3}{6} + O(h^4)$$
$$+ O(\tau^4) + O(h\,\tau^3) + O(h^2\,\tau^2) + O(h^3\,\tau)$$

```
LU_h := 1/τ (subs(p = 0, q = 1, result) − subs(p = 0, q = 0, result) + r/2 · (subs(p = 1, q = 0, result)
    − subs(p = −1, q = 0, result)) − Q·(subs(p = 1, q = 0, result) − 2·subs(p = 0, q = 0, result)
    + subs(p = −1, q = 0, result))) :
 LU_h := expand(simplify(subs(r = a·τ/h, LU_h))):
```

$$LU_h := \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right) a\,h^2}{6} + \frac{\tau^2\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)}{6} - \frac{Q\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right) h^2}{\tau} \tag{2}$$
$$+ \frac{\tau\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)}{2} + a\left(\frac{\partial}{\partial x}\,u(x,t)\right) + \frac{\partial}{\partial t}\,u(x,t)$$

```
f_h := diff(u(x, t), t) + a·diff(u(x, t), x);
```

$$f_h := a\left(\frac{\partial}{\partial x}\,u(x,t)\right) + \frac{\partial}{\partial t}\,u(x,t) \tag{3}$$

```
ξ := LU_h − f_h;
```

$$\xi := \frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right) a\,h^2}{6} + \frac{\tau^2\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)}{6} - \frac{Q\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right) h^2}{\tau} + \frac{\tau\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)}{2} \tag{4}$$

```
ξ + O(h^3) + O(τ^3)
```

$$\frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right) a\,h^2}{6} + \frac{\tau^2\left(\frac{\partial^3}{\partial t^3}\,u(x,t)\right)}{6} - \frac{Q\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right) h^2}{\tau} + \frac{\tau\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)}{2} \tag{5}$$

$$+ O(h^3) + O(\tau^3)$$

$$\xi := subs\left(\frac{Q \cdot diff(u(x,t), x\$2) \cdot h^2}{\tau} = \frac{Q \cdot a \cdot h}{r} \cdot diff(u(x,t), x\$2), \xi \right);$$

$$\xi := \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t) \right) a h^2}{6} + \frac{\tau^2 \left(\frac{\partial^3}{\partial t^3} u(x,t) \right)}{6} - \frac{Q a h \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{r} + \frac{\tau \left(\frac{\partial^2}{\partial t^2} u(x,t) \right)}{2} \quad (6)$$

$$\xi := collect(\xi, [\mathrm{tau}, h], simplify) :$$

$$\xi + O(h^3) + O(\tau^3)$$

$$\frac{\left(\frac{\partial^3}{\partial x^3} u(x,t) \right) a h^2}{6} + \frac{\tau^2 \left(\frac{\partial^3}{\partial t^3} u(x,t) \right)}{6} - \frac{Q a h \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{r} + \frac{\tau \left(\frac{\partial^2}{\partial t^2} u(x,t) \right)}{2} \quad (7)$$

$$+ O(h^3) + O(\tau^3)$$

$$new_\xi := subs($$
$$diff(u(x,t), t\$3) = -a^3 \cdot diff(u(x,t), x\$3),$$
$$diff(u(x,t), t\$2, x) = a^2 \cdot diff(u(x,t), x\$3),$$
$$diff(u(x,t), t, x\$2) = -a \cdot diff(u(x,t), x\$3),$$
$$diff(u(x,t), t\$2) = a^2 \cdot diff(u(x,t), x\$2),$$
$$diff(u(x,t), x, t) = -a \cdot diff(u(x,t), x\$2),$$
$$diff(u(x,t), t) = -a \cdot diff(u(x,t), x),$$
$$\xi$$
$$) :$$

$$new_\xi := collect(new_\xi, [\mathrm{tau}, h], simplify);$$

$$new_\xi := \frac{\left(\frac{\partial^3}{\partial x^3} u(x,t) \right) a h^2}{6} - \frac{\tau^2 a^3 \left(\frac{\partial^3}{\partial x^3} u(x,t) \right)}{6} - \frac{Q a h \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{r} \quad (8)$$
$$+ \frac{\tau a^2 \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{2}$$

$$subs\left(\tau = \frac{r \cdot h}{a}, new_\xi \right)$$

$$\frac{\left(\frac{\partial^3}{\partial x^3} u(x,t) \right) a h^2}{6} - \frac{r^2 h^2 a \left(\frac{\partial^3}{\partial x^3} u(x,t) \right)}{6} - \frac{Q a h \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{r} \quad (9)$$
$$+ \frac{r h a \left(\frac{\partial^2}{\partial x^2} u(x,t) \right)}{2}$$

$$new_\xi + O(h^3) + O(\tau^3)$$

$$\frac{\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right) a\,h^2}{6} - \frac{\tau^2\,a^3\left(\frac{\partial^3}{\partial x^3}\,u(x,t)\right)}{6} - \frac{Q\,a\,h\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{r} + \frac{\tau\,a^2\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{2} \quad (10)$$

$$+ O(h^3) + O(\tau^3)$$

$k := 1; m := 1;$

$$k := 1$$

$$m := 1 \quad (11)$$

#Г-форма первого дифференциального приближения

$g_1 := -\left(coeff\left(coeff\left(\xi, \text{tau}, 1\right), h, 0\right)\cdot\tau + coeff\left(coeff\left(\xi, \text{tau}, 0\right), h, 1\right)\cdot h\right):$

$f_h = g_1$

$$a\left(\frac{\partial}{\partial x}\,u(x,t)\right) + \frac{\partial}{\partial t}\,u(x,t) = -\frac{\tau\left(\frac{\partial^2}{\partial t^2}\,u(x,t)\right)}{2} + \frac{Q\,a\,h\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{r} \quad (12)$$

#П-форма первого дифференциального приближения

$p_1 := subs\left(diff\left(u(x,t), t\$2\right) = a^2\cdot diff\left(u(x,t), x\$2\right), a\cdot diff\left(u(x,t), t, x\right) = -a^2\cdot diff\left(u(x,t), x\$2\right), \tau = \frac{r\cdot h}{a}, g_1\right)$

$$p_1 := -\frac{r\,h\,a\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{2} + \frac{Q\,a\,h\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{r} \quad (13)$$

$f_h = p_1$

$$a\left(\frac{\partial}{\partial x}\,u(x,t)\right) + \frac{\partial}{\partial t}\,u(x,t) = -\frac{r\,h\,a\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{2} + \frac{Q\,a\,h\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)}{r} \quad (14)$$

$simplify(coeff(p_1, h, 1))\cdot h$

$$\frac{a\left(\frac{\partial^2}{\partial x^2}\,u(x,t)\right)\left(-r^2 + 2\,Q\right) h}{2\,r} \quad (15)$$

#Порядок аппроксимации

$_p := 1$

$$_p := 1 \quad (16)$$

$f := \lambda^s\cdot e^{i\cdot p\cdot\alpha}$

$$f := \lambda^s\, e^{i\,p\,\alpha} \quad (17)$$

$eq := subs(s=1,p=0,f) - subs(s=0,p=0,f) + \frac{r}{2}\cdot(subs(s=0,p=1,f) - subs(s=0,p=-1,f)) - Q\cdot(subs(s=0,p=1,f) - 2\cdot subs(s=0,p=0,f) + subs(s=0,p=-1,f)) = 0$

$$eq := \lambda e^{0} - e^{0} + \frac{r\left(e^{i\alpha} - e^{-i\alpha}\right)}{2} - Q\left(e^{i\alpha} - 2e^{0} + e^{-i\alpha}\right) = 0 \quad (18)$$

$solution := solve(eq, \lambda);$

$$solution := 1 - \frac{r e^{i\alpha}}{2} + \frac{r e^{-i\alpha}}{2} + Q e^{i\alpha} + Q e^{-i\alpha} - 2Q \quad (19)$$

$sol2 := simplify(subs(e^{i\cdot\alpha} = \cos(\alpha) + i\cdot\sin(\alpha), e^{-i\cdot\alpha} = \cos(\alpha) - i\cdot\sin(\alpha), solution));$

$$sol2 := -\sin(\alpha)\, i\, r + 2\cos(\alpha)\, Q - 2Q + 1 \quad (20)$$

$\lambda_2 := simplify(coeff(sol2, i, 1)^2 + (sol2 - coeff(sol2, i, 1)\cdot i)^2)$

$$\lambda_2 := \sin(\alpha)^2 r^2 + (2\cos(\alpha)\, Q - 2Q + 1)^2 \quad (21)$$

$\lambda_2 := expand(subs(\cos(\alpha) = c, \sin(\alpha)^2 = 1 - c^2, \lambda_2))$

$$\lambda_2 := 4Q^2c^2 - c^2r^2 - 8Q^2c + 4Q^2 + 4cQ + r^2 - 4Q + 1 \quad (22)$$

#условие можно свести к λ_2-1<=0
#можно вынести (1-c) и получить:
$(1-c)\cdot(2\cdot r^2 - 4\cdot Q + (4Q^2 - r^2)(1-c))$

$$(1-c)(2r^2 - 4Q + (4Q^2 - r^2)(1-c)) \quad (23)$$

#(1-c)>=0 при -1<c<1. Поэтому $(2\cdot r^2 - 4\cdot Q + (4Q^2 - r^2)(1-c)) \le 0$
$f_1 := 2\cdot r^2 - 4\cdot Q + (4Q^2 - r^2)(1-c):$
$f_1 := collect(f_1, c, simplify);$

$$f_1 := (-4Q^2 + r^2)c + 4Q^2 + r^2 - 4Q \quad (24)$$

#f_1 линейна относительно c, поэтому максимум будет находится на границе отрезка с [-1, 1]
$eq_1 := subs(c=1, f_1) \le 0$

$$eq_1 := 2r^2 - 4Q \le 0 \quad (25)$$

$solve(eq_1, Q)$

$$\left\{\frac{r^2}{2} \le Q\right\} \quad (26)$$

$eq_2 := subs(c=-1, f_1) \le 0$

$$eq_2 := 8Q^2 - 4Q \le 0 \quad (27)$$

$solve(eq_2, Q)$

$$\left[0, \frac{1}{2}\right] \quad (28)$$

#то есть $\frac{r^2}{2} \le Q \le 1\backslash 2$ *— область устойчивости*

$$solve\left(\frac{d}{d\,c}(\lambda_2)=0,\ c\right)$$

$$\frac{2\,Q\,(2\,Q-1)}{4\,Q^2-r^2} \tag{29}$$

$$subs(c=1,\ \lambda_2)$$

$$1 \tag{30}$$

$$simplify(subs(c=-1,\ \lambda_2))$$

$$16\left(Q-\frac{1}{4}\right)^2 \tag{31}$$

#Худшей гармоникой будет максимум из предыдуших трех значений.Но первое значение меньше 0, поэтому худшая гармоника будет при cos(α)=-+1